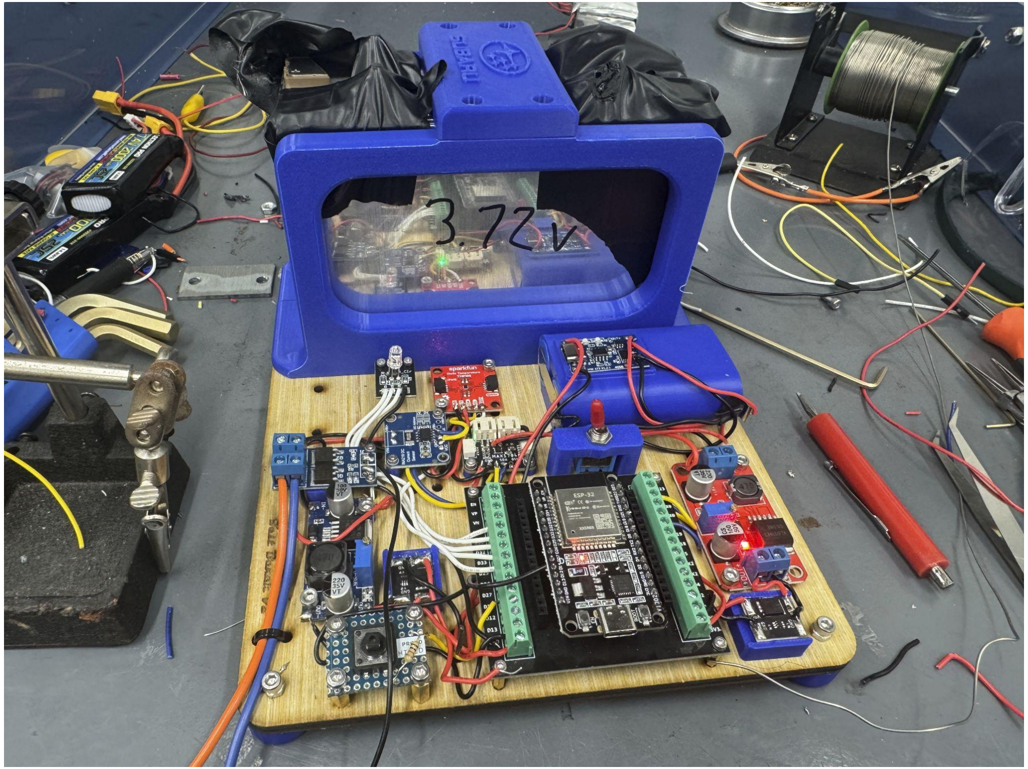
Subaru Internship End of Year Presentation

By Chase Quijano

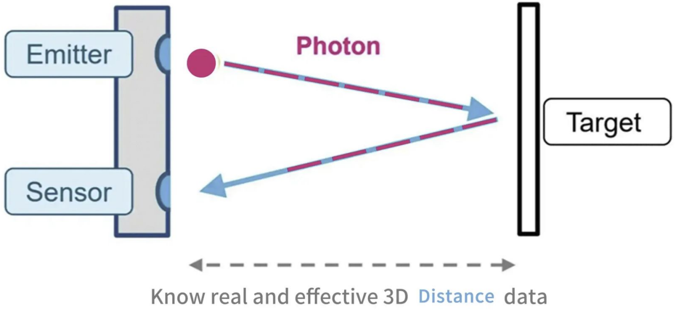
Bug Box Test Bench

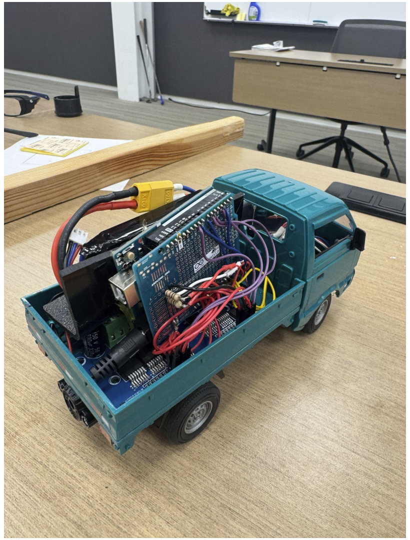
- Developed to be the groundwork for the bugbox's development
- ESP8266 microcontroller
- Switching Mosfet
- Current Sensor
- Temperature Sensor
- Interactive Pushbutton
- TP4056 lipo charger
- 18650 battery pack
- Selective diode switching between two power sources



Autonomous Car

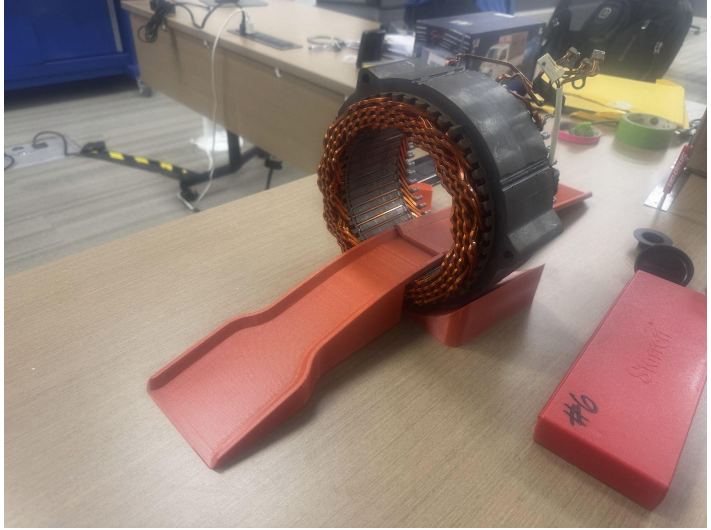
- Developed for golf themed event
- Modified RC K-Truck chassis
- Arduino Uno microcontroller
- Time of Flight (TOF) sensors utilized to determine distance of objects from the front, left, and right and determine tire rotation and speed
- DRV8322 motor driver
- Custom Subaru cargo container

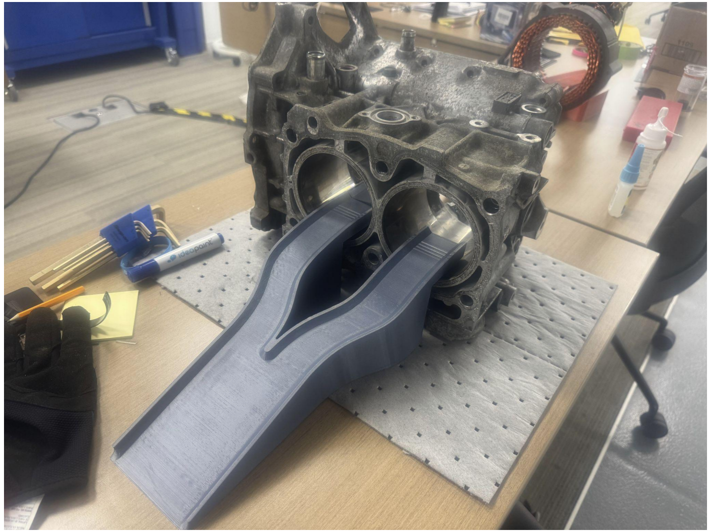




Golf Ramps

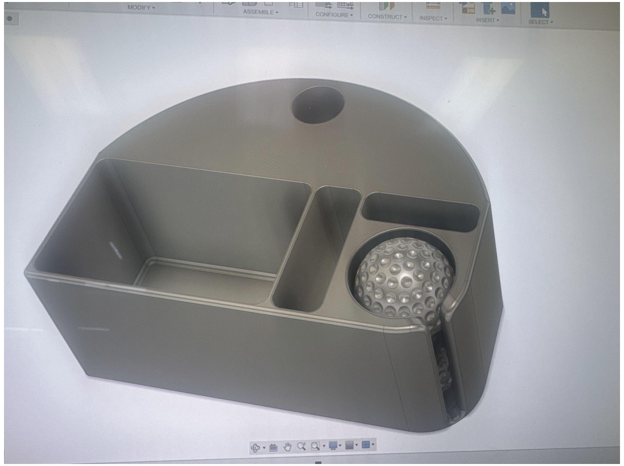
- Developed for golf themed event
- 3D printed with modularity in mind
- Consists of embedded magnets to interlock sections of ramps
- Consists of weights to keep fixed in place during puts
- V ramp design allows for more than one route during golfing

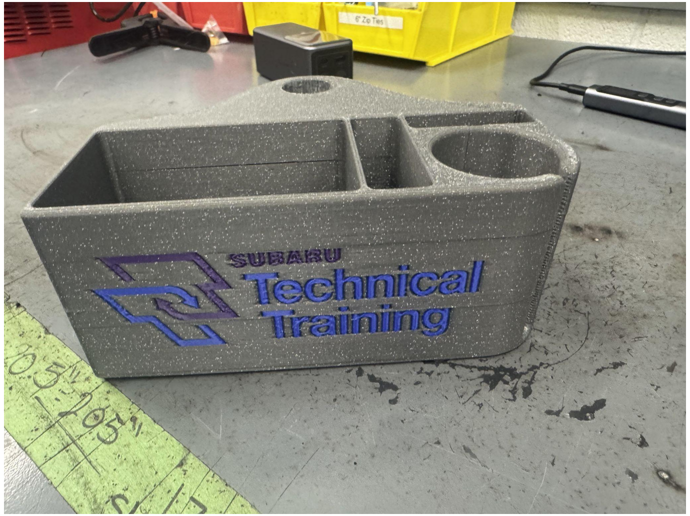




Golf Caddy

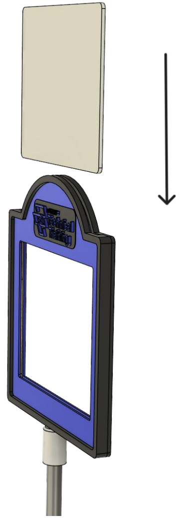
- Designed for golf themed event
- Mounts onto pole
- Pencil holding section for writing on cards
- Card section for ease of reach
- Golf ball hopper with protruding tab, allowing for accessing lower golf balls
- Multi Color print done with Bambu Lab AMS





Event Signs

- Mounts onto pole
- Custom laser cut white board is easily removable and can be written on for different events
- Clamshell design with several layers of laser cut and 3D printed components

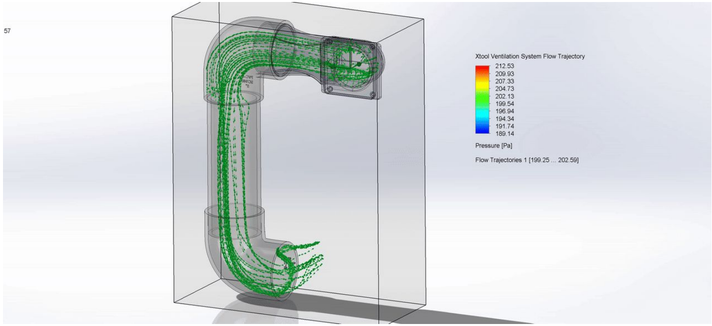


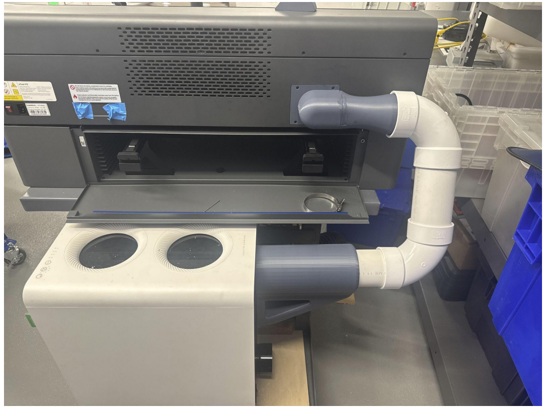
Bug boxes

- Designed as a multi node network that allows remote activation and deactivation of electrical components inside vehicles for technical training and competition purposes
- 2 18650 Battery (6000 mAh)
- Xiao ESP32 S3 microcontroller
- Passive cooling
- Transistor rated for high voltage
- 3D printed enclosure, easy to disassemble with four points of fastening

Xtool Laser cutter modifications and organizational system

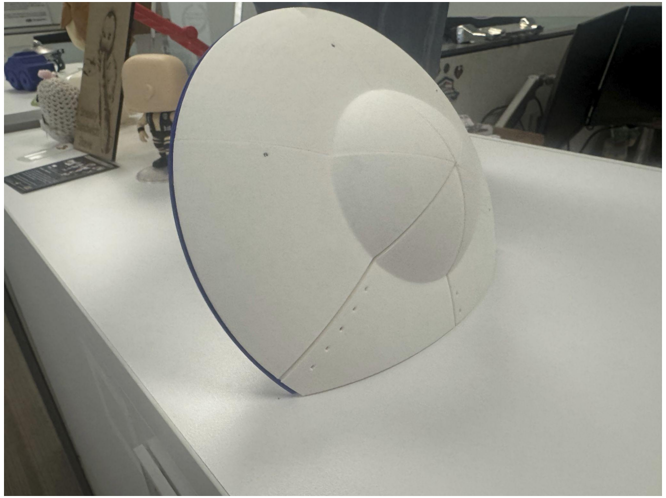
- Developed to make Xtool laser cutter easier to use
- Ventilation system developed to be compact and note require additional setup
- Organizational system allows for easy access to materials during cutting
- Xtool laser modified to permit conveyor belt cutting and regular cutting without equipment changes

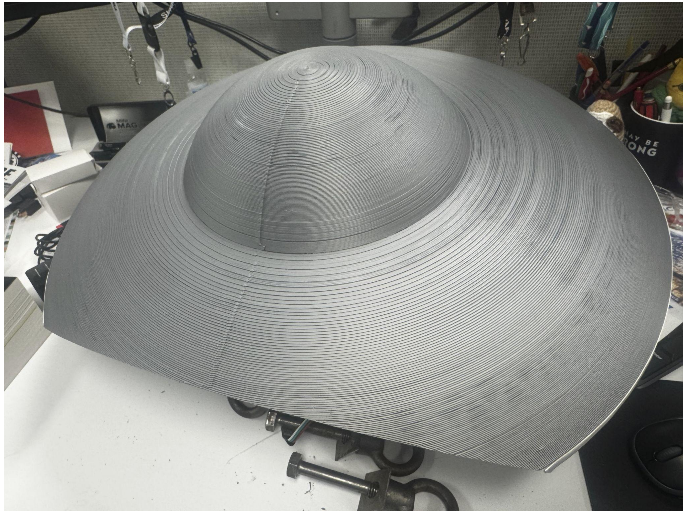




Large Scale UFO

- Designed to be 3D printed in multiple sections and riveted together
- Eventually finalized design was a singular component at a smaller scale to minimize printing costs
- RGB strip revolving around the rim for visual effects





3D Printed TH2B

- 3D printed TH2Bs made to be sent out to training centers for demonstrative purposes
- Dremeling out and pockets in order to fit necessary tolerances needed for easy assembly and disassembly

Component Organizational System

- Developed to have a centralized location for all small electronics
- Comprised of a 3D printed parametric box system
- Two layers of organization with sliding for access to different sections
- Section for labeling designed into print